

KAI TOYOSAWA

Last updated: September 2026

Mathematik Münster
Universität Münster
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Academic Employment

Postdoctoral researcher, University of Münster.
Mentor: Prof. David Kerr

October 2025 - Present.

Education

Ph.D. in Mathematics, Vanderbilt University.
Advisor: Prof. Jesse Peterson

May 2025.

M.A. in Mathematics, University of California, Los Angeles.

June 2019.

B.S. in Mathematics, University of California, Los Angeles.

June 2019.

Publications & Preprints

1. K. Toyosawa: *Weak exactness and amalgamated free product of von Neumann algebras*, J. Funct. Anal. **288** (2025), no. 4, 110759.
2. K. Toyosawa, Z. Yang: *On relative biexactness of amalgamated free product von Neumann algebras*, arXiv:2505.19508.
3. I. Chifan, K. Toyosawa, Z. Yang: *Relative Biexactness for Relative Hyperbolic Groups and Some Applications*, arXiv:2608.23991.
4. A. Fernández Quero, K. Toyosawa: *Properly proximal countable measured groupoids*, arXiv:2609.01253.
5. D. Osin, K. Toyosawa, Z. Yang: *A paradoxical route from hyperbolic geometry to proper proximality*, arXiv:2609.21896.

Honors & Awards

B.F. Bryant Teaching Award, Vanderbilt University, 2024.
Richard F. Arenstorf Fellowship, Vanderbilt University, 2019 - 2022.

Seminars & Conferences

Conference Presentations

The CJK Young Researchers in Functional Analysis 2026, Seoul National University, August 2026.
Young Mathematicians in C^* -Algebras (YMC * A), Oxford University, August 2026.
Great Plains Operator Theory Symposium (GPOTS), University of Iowa, May 2026.

NSF/CBMS Regional Conference in the Mathematical Sciences: Classifying amenable operator algebras, Texas Christian University, June 2025

JMM AMS Special Session on Advances in Operator Algebras, Seattle, January 2025.

Young Mathematicians in C^* -Algebras (YMC * A), University of Glasgow, August 2024.

Great Plains Operator Theory Symposium (GPOTS), University of Nebraska–Lincoln, June 2024.

Invited Seminars

Operator Algebras Seminar, University of Tokyo, March 2026.

Seminar operator algebras, KU Leuven, December 2025.

Senior Seminar C^* -Algebras, University of Münster, November 2025.

Operator Theory Seminar, University of Virginia, January 2025.

Noncommutative Geometry & Functional Analysis Seminar (online), SCMS Fudan University, November 2024.

Operator Theory Seminar, University of Iowa, November 2024.

Functional Analysis Seminar, Purdue University, October 2024.

Functional Analysis Seminar, University of California, San Diego, October 2024.

Subfactor Seminar, Vanderbilt University, February 2024.